

NIER AGDM, AB, Canada

WMO: 714950

Lat: 51.3694N

Lon: 114.0992W

Elev: 3757

StdP: 12.81

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.7	-12.7	-24.5	1.6	-18.2	-18.1	2.3	-12.2	29.4	30.9	26.7	25.9	7.3	310	0.731

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	81.5	60.5	78.1	59.5	74.9	58.3	64.4	75.5	62.2	73.5	60.2	71.4	9.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	89.5	70.5	57.7	81.7	67.9	55.6	75.7	65.0	31.6	75.6	29.9	73.9	28.3	71.6	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.7	20.7	-25.9	88.3	4.1	4.1	-28.9	91.2	-31.3	93.6	-33.6	95.9	-36.5	98.9
			-26.6	69.0	4.1	2.6	-29.5	70.8	-31.9	72.3	-34.1	73.8	-37.1	75.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	22.1	18.5	28.3	37.7	48.6	54.7	59.9	58.2	50.9	40.1	27.7	19.2
	DBStd	18.34	14.54	15.36	13.40	9.32	7.85	5.44	5.12	6.06	8.84	9.39	13.41	14.43
	HDD50	4994	864	882	674	378	123	16	1	8	96	326	670	956
	HDD65	9541	1329	1302	1139	818	509	310	172	223	427	773	1118	1421
	CDD50	960	0	0	1	10	79	158	308	261	123	18	2	0
	CDD65	32	0	0	0	0	1	2	14	11	4	0	0	0
	CDH74	872	0	0	0	4	41	55	314	319	134	5	0	0
	CDH80	160	0	0	0	0	2	5	56	75	22	0	0	0
	WSAvg	9.8	11.4	9.6	9.9	10.6	10.7	9.9	8.3	7.8	8.7	9.8	10.4	10.6
	PrecAvg	18.6	0.6	0.4	0.7	1.2	2.7	4.0	2.7	2.5	1.7	0.8	0.7	0.6
(15) Precipitation	PrecMax	25.0	1.1	1.0	2.1	3.0	5.0	11.7	6.1	4.6	4.2	1.7	2.0	1.5
	PrecMin	12.5	0.0	0.0	0.2	0.4	1.0	1.1	0.9	0.5	0.2	0.2	0.1	0.1
	PrecStd	2.6	0.3	0.3	0.4	0.7	1.1	2.1	1.4	1.1	0.9	0.4	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	53.9	59.3	70.3	78.7	79.8	85.0	86.4	82.9	70.7	60.2	51.5
		MCWB	39.2	40.1	42.9	48.3	52.8	61.6	63.0	60.2	56.6	50.4	43.6	37.5
	2%	DB	46.4	46.7	54.3	64.6	73.8	74.9	81.2	81.5	78.1	64.4	53.5	45.2
		MCWB	36.8	36.3	40.6	46.3	52.4	58.1	63.0	60.2	55.8	48.1	41.3	36.1
	5%	DB	43.5	42.5	49.6	59.1	69.5	71.5	77.4	77.3	73.3	59.2	49.0	41.9
		MCWB	35.1	34.1	37.9	43.2	50.6	56.0	62.3	59.1	54.4	45.4	38.6	34.2
	10%	DB	41.0	38.4	45.6	53.6	65.2	68.0	74.0	73.1	67.9	55.1	44.6	38.4
		MCWB	33.6	31.5	36.2	40.4	48.5	54.6	60.8	58.6	52.7	43.2	36.0	31.5
	0.4%	WB	39.9	40.6	44.5	49.5	55.8	64.3	68.9	65.5	59.1	52.2	45.7	39.5
		MCDB	49.1	52.6	56.0	68.7	72.6	74.4	78.9	77.1	76.2	67.6	59.0	47.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	36.9	41.4	46.7	53.1	60.4	65.6	63.0	57.2	49.1	41.9	36.8
		MCDB	45.6	46.1	52.6	63.5	70.2	71.4	76.8	75.2	73.6	62.1	52.5	44.6
	5%	WB	35.6	34.2	38.7	43.9	51.3	58.0	63.4	60.9	55.3	46.3	39.3	34.3
		MCDB	43.0	42.0	48.2	57.4	66.9	68.1	74.2	73.7	70.4	58.1	48.3	41.7
	10%	WB	33.6	31.6	36.4	41.1	49.5	55.9	61.5	58.9	53.1	43.9	36.5	31.6
		MCDB	40.7	37.8	45.2	51.6	63.2	65.5	72.0	71.3	66.9	53.8	43.9	38.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	19.1	18.9	20.8	22.5	20.7	24.1	24.3	23.2	21.0	18.5	17.2
		MCDBR	18.8	22.6	25.1	30.1	29.6	27.7	30.0	32.1	32.4	26.5	23.0	18.9
		MCWBR	12.0	14.9	14.5	15.5	13.5	13.8	15.7	14.8	14.1	14.1	13.3	12.7
	5% WB	MCDBR	18.7	22.1	23.6	28.1	26.8	23.9	26.8	28.4	29.5	25.4	22.8	18.8
		MCWBR	12.4	14.7	14.4	14.9	12.8	13.1	15.4	14.7	13.7	13.9	13.8	13.0
(40) Clear-Sky Solar Irradiance	taub		0.248	0.263	0.276	0.328	0.336	0.341	0.337	0.351	0.307	0.275	0.249	0.234
	taud		2.410	2.445	2.483	2.346	2.385	2.409	2.419	2.390	2.479	2.553	2.498	2.395
	Ebn at Noon		254	282	298	289	291	290	289	279	282	274	255	242
	Edh at Noon		18	23	27	35	36	35	34	34	27	21	16	16
(44) All-Sky Solar Radiation	RadAvg		354	665	1077	1468	1741	1850	2022	1659	1172	721	386	277
	RadStd		23	50	77	109	133	134	122	116	130	76	27	21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page